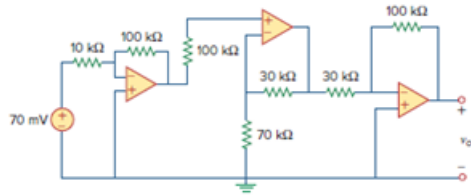


Problem

Find v_o in the op amp circuit of Fig. 5.92.

Figure 5.92:



Step-by-step solution

Step 1 of 4

Refer to Figure 5.92 in the textbook.

Draw the following circuit:

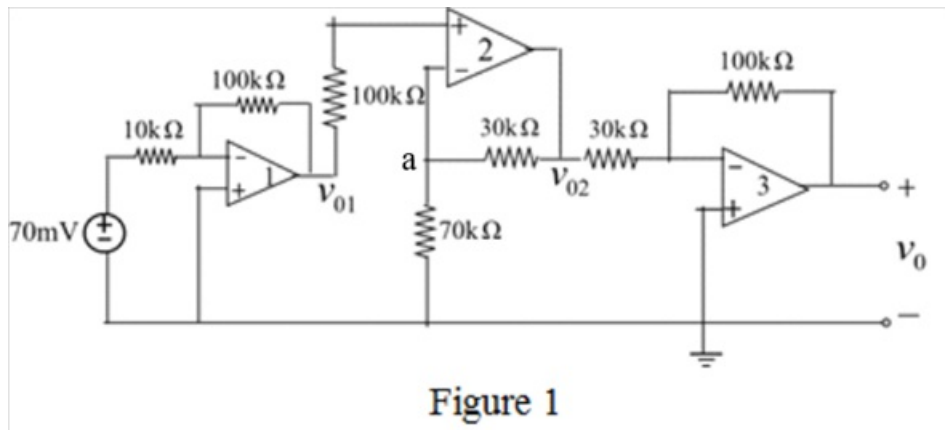


Figure 1

Step 2 of 4

The op-amp 1 shown in Figure 1 is an inverting amplifier.

Hence, the output of the op-amp 1 is

$$v_{o1} = -\left(\frac{R_f}{R}\right)v_i$$

Substitute $100\text{ k}\Omega$ for R_f , $10\text{ k}\Omega$ for R , and 70 mV for v_i .

$$\begin{aligned} v_{o1} &= -\left(\frac{100\text{ k}\Omega}{10\text{ k}\Omega}\right)(70\text{ mV}) \\ &= -700\text{ mV} \\ &= -0.7\text{ V} \end{aligned}$$

Step 3 of 4

The op-amp 2 shown in Figure 1 is a non-inverting amplifier.

The voltage v_{01} is the input to the non-inverting amplifier. Since, the voltage drop across the $100\text{ k}\Omega$ resistor is zero, as no current flows into the terminal of the ideal op amp.

Calculate the voltage at node **a**.

Apply voltage divider rule at node **a**.

$$\begin{aligned}v_a &= \frac{70\text{k}}{70\text{k} + 30\text{k}} v_{01} \\&= \frac{70}{100} (-0.7) \\&= 0.49\text{ V}\end{aligned}$$

Write the expression for the output of the op-amp 2.

$$v_{02} = \left(1 + \frac{R_f}{R}\right) v_a$$

Substitute $30\text{ k}\Omega$ for R_f , $100\text{ k}\Omega$ for R , and -0.49 V for v_a .

$$\begin{aligned}v_{02} &= \left(1 + \frac{30\text{k}}{100\text{k}}\right) (-0.49) \\&= \left(\frac{130}{100}\right) (-0.49) \\&= -0.637\text{ V}\end{aligned}$$

Step 4 of 4

The op-amp 3 is an inverting amplifier.

The voltage v_{02} is the input to the inverting amplifier.

Hence, the output of the op-amp 3 is,

$$v_o = -\left(\frac{R_f}{R}\right) v_{02}$$

Substitute $100\text{ k}\Omega$ for R_f , $30\text{ k}\Omega$ for R , and -0.637 V for v_{02} .

$$\begin{aligned}v_o &= -\left(\frac{100\text{k}}{30\text{k}}\right) (-0.637\text{ V}) \\&= 2.12\text{ V} \\&\approx 2\text{ V}\end{aligned}$$

Therefore, the value of output voltage v_o is 2V.